



**BAULKHAM HILLS
HIGH
SCHOOL**

2020

**YEAR 12
HALF YEARLY
ASSESSMENTS**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black pen
- Write your NESA# and Teacher's name on your answer booklet
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks: 53

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

Section II – 48 marks (pages 4 – 7)

- Attempt Questions 6 – 8
- Allow about 1 hour and 22 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5

1 $\int \frac{dx}{4 + 9x^2} = ?$

(A) $\frac{1}{3} \tan^{-1}\left(\frac{3x}{2}\right) + c$

(B) $\frac{1}{6} \tan^{-1}\left(\frac{3x}{2}\right) + c$

(C) $\frac{1}{3} \tan^{-1}\left(\frac{2x}{3}\right) + c$

(D) $\frac{1}{6} \tan^{-1}\left(\frac{2x}{3}\right) + c$

- 2 The region bounded by the curve $y = x^4$, the y -axis and the line $y = 16$ is rotated about the y -axis.

Which of the following expressions is correct for the volume of the solid of revolution created by this rotation?

(A) $\pi \int_0^2 x^8 dx$

(B) $\pi \int_0^{16} x^8 dx$

(C) $\pi \int_0^2 \sqrt{y} dy$

(D) $\pi \int_0^{16} \sqrt{y} dy$

3 What is the maximum value of $6\cos\theta + 4\sin\theta$?

(A) 10

(B) $2\sqrt{13}$

(C) 6

(D) $2\sqrt{5}$

4 Which of the following is the primitive of $\sin^2 x$?

(A) $x - \frac{1}{2} \sin 2x + c$

(B) $x + \frac{1}{2} \sin 2x + c$

(C) $\frac{x}{2} - \frac{1}{4} \sin 2x + c$

(D) $\frac{x}{2} + \frac{1}{4} \sin 2x + c$

5 Using the substitution $u = 2x + 1$, $\int (2x - 1)\sqrt{2x + 1} \, dx$ can be expressed as

(A) $\frac{1}{2} \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}} \right) du$

(B) $\frac{1}{2} \int \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$

(C) $2 \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}} \right) du$

(D) $2 \int \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$

END OF SECTION I

Section II

48 marks

Attempt Questions 6 – 8

Allow about 1 hour 22 minutes for this section

Answer each question on the appropriate answer sheet. Each answer sheet must show your NES A#. Extra paper is available.

In Questions 6 to 8, your responses should include relevant mathematical reasoning and/or calculations.

Marks

Question 6 (15 marks) Use a *separate* answer sheet

(a) Differentiate the following functions with respect to x

(i) $f(x) = [\sin^{-1}(3x)]^3$ 2

(ii) $f(x) = \ln(\cos^{-1}x)$ 2

(iii) $f(x) = (x^2 + 1)\tan^{-1}x$ 2

(b) Evaluate

(i) $\int_{-2}^2 \frac{dx}{x^2 + 4}$ 2

(ii) $\int_0^1 \frac{dx}{\sqrt{4 - 3x^2}}$ 3

(c) Solve $2\sin\theta - \cos\theta = 1$, for $0^\circ \leq \theta \leq 360^\circ$, correct to the nearest 4 minute, using the substitution $t = \tan \frac{\theta}{2}$

Question 7 (20 marks) Use a *separate* answer sheet

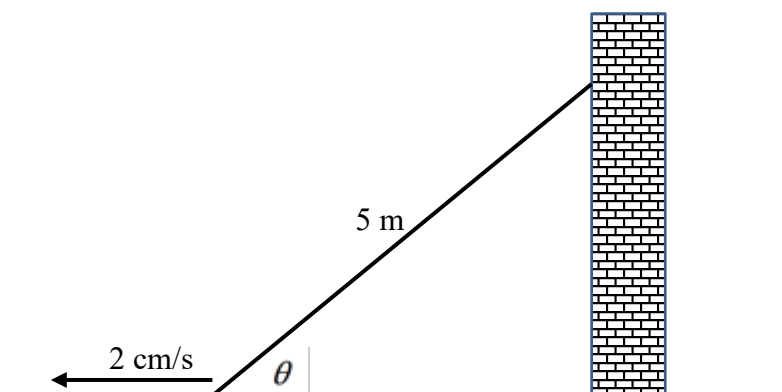
(a) Find the following integrals using the given substitution

(i) $\int x(1 - x^2)^5 dx$ using the substitution $u = 1 - x^2$ 2

(ii) $\int \frac{dx}{2x \ln x}$ using the substitution $u = \ln x$ 3

(iii) $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1 + \tan x}} dx$ using the substitution $u = 1 + \tan x$ 3

(b) A 5 metre ladder is leaning against a wall on level ground. The base of the ladder is sliding away from the wall at a rate of 2 cm/s. 3



Find the rate at which the angle of inclination the ladder makes with the ground, θ , is changing when the foot of the ladder is 3 metres from the base of the wall.

Question 7 continues on page 6

Question 7 (continued)

- (c) (i) Express $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$ 2

- (ii) Hence, or otherwise, solve; 3

$$\cos x - \sqrt{3}\sin x = 1 \quad \text{for} \quad 0 \leq x \leq 2\pi$$

- (d) (i) Show that $\frac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$ 2

- (ii) Hence find $\int \sec^4 x \, dx$ 2

End of Question 7

Question 8 (13 marks) Use a *separate* answer sheet

- (a) Solve; 4

$$\sin 3x = \sin 2x \cos x \quad \text{for} \quad 0 \leq x \leq \frac{\pi}{2}$$

- (b) Find the volume of the solid generated when the curve $y = \cos x + \sec x$ is rotated about the x -axis between $x = 0$ and $x = \frac{\pi}{3}$ 4

- (c) (i) Show that; 2

$$\frac{d}{dx} \left(\tan^{-1} \frac{1-x}{1+x} \right) = -\frac{1}{1+x^2}$$

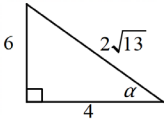
- (ii) Prove that for $x > -1$, 2

$$\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4}$$

- (iii) Hence calculate $\tan^{-1}(\sqrt{2} - 1)$ 1

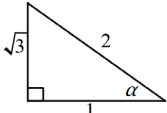
End of paper

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 EXTENSION HALF YEARLY 2020 SOLUTIONS

Solution		Marks	Comments
SECTION I			
1. B -	$\int \frac{dx}{4 + 9x^2} = \frac{1}{3} \int \frac{3dx}{2^2 + (3x)^2}$ $= \frac{1}{3} \times \frac{1}{2} \tan^{-1} \left(\frac{3x}{2} \right) + c$ $= \frac{1}{6} \tan^{-1} \left(\frac{3x}{2} \right) + c$	1	
2. D -	$y = x^4 \Rightarrow x = y^{\frac{1}{4}} \quad V = \pi \int_0^{16} \left(y^{\frac{1}{4}} \right)^2 dy$ $= \pi \int_0^{16} y^{\frac{1}{2}} dy$ $= \pi \int_0^{16} \sqrt{y} dy$	1	
3. B -	$6\cos\theta + 4\sin\theta$ $= 2\sqrt{13} \sin(\theta + \alpha)$ $\therefore$ maximum value is $2\sqrt{13}$ 	1	
4. C -	$f'(x) = \sin^2 x$ $= \frac{1}{2}(1 - \cos 2x)$ $f(x) = \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) + c$ $= \frac{x}{2} - \frac{1}{4} \sin 2x + c$	1	
5. B -	$\int (2x - 1) \sqrt{2x + 1} du$ $= \frac{1}{2} \int (u - 2) \sqrt{u} du$ $= \frac{1}{2} \int \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$ $u = 2x + 1$ $du = 2dx$	1	

SECTION II		
Solution	Marks	Comments
QUESTION 6		
6(a) (i) $f(x) = [\sin^{-1}(3x)]^3$ $f'(x) = 3[\sin^{-1}(3x)]^2 \times \frac{3}{\sqrt{1-9x^2}}$ $= \frac{9[\sin^{-1}(3x)]^2}{\sqrt{1-9x^2}}$	2	2 marks • Correct solution 1 mark • Correctly differentiates $\sin^{-1}(3x)$ • Uses the chain rule with a valid attempt to differentiate $\sin^{-1}(3x)$
6(a) (ii) $f(x) = \ln(\cos^{-1}x)$ $f'(x) = \frac{\left(-\frac{1}{\sqrt{1-x^2}}\right)}{\cos^{-1}x}$ $= -\frac{1}{\sqrt{1-x^2} \cos^{-1}x}$	2	2 marks • Correct solution 1 mark • Correctly differentiates $\cos^{-1}x$ • Differentiates the log function with a valid attempt to differentiate $\cos^{-1}x$
6(a) (iii) $f(x) = (x^2 + 1)\tan^{-1}x$ $f'(x) = (x^2 + 1)\left(\frac{1}{1+x^2}\right) + (\tan^{-1}x)(2x)$ $= 1 + 2x\tan^{-1}x$	2	2 marks • Correct solution 1 mark • Correctly differentiates $\tan^{-1}x$ • Uses the product rule with a valid attempt to differentiate $\tan^{-1}x$
6(b) (i) $\int_{-2}^2 \frac{dx}{x^2 + 4} = 2 \times \frac{1}{2} \left[\tan^{-1}\left(\frac{x}{2}\right) \right]_0^2$ $= \tan^{-1} 1 - \tan^{-1} 0$ $= \frac{\pi}{4} - 0$ $= \frac{\pi}{4}$	2	2 marks • Correct solution 1 mark • Finds the correct primitive
6(b) (ii) $\int_0^1 \frac{dx}{\sqrt{4-3x^2}} = \frac{1}{\sqrt{3}} \int_0^1 \frac{\sqrt{3} dx}{\sqrt{4-3x^2}}$ $= \frac{1}{\sqrt{3}} \left[\sin^{-1} \frac{\sqrt{3}x}{2} \right]_0^1$ $= \frac{1}{\sqrt{3}} \left(\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \right)$ $= \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - 0 \right)$ $= \frac{\pi}{3\sqrt{3}}$	3	3 marks • Correct solution 2 marks • Finds the correct primitive • Correctly evaluates definite integral using a primitive in the form of $k \sin^{-1} \frac{\sqrt{3}x}{2}$ 1 mark • Establishes the primitive as $k \sin^{-1} \frac{\sqrt{3}x}{2}$

Solution		Marks	Comments
6(c) $2\sin\theta - \cos\theta = 1$ $2\left(\frac{2t}{1+t^2}\right) - \left(\frac{1-t^2}{1+t^2}\right) = 1$ $4t - 1 + t^2 = 1 + t^2$ $4t - 2 = 0$ $t = \frac{1}{2}$ $\tan\frac{\theta}{2} = \frac{1}{2}$ $\frac{\theta}{2} = 26^\circ 34'$ $\theta = 53^\circ 8'$ $\theta = 53^\circ 8', 180^\circ$		4	4 marks • Correct solution 3 marks • Correctly finds the solution $\theta = 53^\circ 8'$ • Transforms the trig equation into a polynomial equation and establishes $\theta = 180^\circ$ 2 marks • Transforms the trig equation into a polynomial equation 1 mark • Substitutes the correct t -results for $\sin\theta$ and $\cos\theta$ • establishes $\theta = 180^\circ$ as a solution
QUESTION 7			
7(a) (i) $\int x(1-x)^5 dx = -\frac{1}{2} \int u^5 du$ $= -\frac{1}{12} u^6 + c$ $= -\frac{1}{12} (1-x^2)^6 + c$		2	2 marks • Correct solution 1 mark • Rewrites integrand in terms of u
7(a) (ii) $\int \frac{dx}{2x \ln x} = \frac{1}{2} \int \frac{du}{u}$ $= \frac{1}{2} \ln u + c$ $= \frac{1}{2} \ln \ln x + c$		3	3 marks • Correct solution 2 marks • Finds the primitive in terms of u 1 mark • Rewrites integrand in terms of u
7(a) (iii) $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1+\tan x}} dx = \int_1^2 \frac{2du}{\sqrt{u}}$ $= 2 \left[\sqrt{u} \right]_1^2$ $= 2(\sqrt{2} - 1)$		3	3 marks • Correct solution 2 marks • Finds the primitive in terms of u • Changes both the integrand and the limits in terms of u 1 mark • Rewrites integrand in terms of u • Expresses limits in terms of u

Solution	Marks	Comments
7(b) $\frac{dx}{dt} = \frac{1}{50} \text{ m/s}$ $\cos \theta = \frac{x}{5}$ $\frac{d\theta}{dt} = \frac{dx}{dt} \times \frac{d\theta}{dx}$ $\theta = \cos^{-1} \frac{x}{5}$ $= \frac{1}{50} \times \frac{-1}{\sqrt{25-x^2}}$ $\frac{d\theta}{dx} = \frac{-1}{\sqrt{25-x^2}}$ $= \frac{-1}{50\sqrt{25-x^2}}$ when $x = 3$; $\frac{d\theta}{dt} = \frac{-1}{50\sqrt{25-9}}$ $= -\frac{1}{200} \text{ rad/s}$ $\therefore$ the angle of inclination is decreasing at a rate of $\frac{1}{200}$ radians/second	3	3 marks • Correct solution, including correct units 2 marks • Finds a correct expression for $\frac{d\theta}{dt}$ 1 mark • Finds a correct expression for $\frac{d\theta}{dx}$ • Uses the chain rule to find an expression for $\frac{d\theta}{dt}$
7 (c) (i) $\cos x - \sqrt{3} \sin x$ $= 2 \cos \left(x + \frac{\pi}{3} \right)$  $\tan \alpha = \sqrt{3}$ $\alpha = \frac{\pi}{3}$	2	2 marks • Correct solution 1 mark • Finds R or α
7(c) (ii) $\cos x - \sqrt{3} \sin x = 1$ $2 \cos \left(x + \frac{\pi}{3} \right) = 1$ $\cos \left(x + \frac{\pi}{3} \right) = \frac{1}{2}$ $x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}$ $x = 0, \frac{4\pi}{3}, 2\pi$	3	3 marks • Finds all three solutions 2 marks • Finds two values of x 1 mark • Finds two answers for $x + \frac{\pi}{3}$ • Finds one value of x
7(d) (i) $\frac{d}{dx}(\tan^3 x) = 3(\tan^2 x)(\sec^2 x)$ $= 3(\sec^2 x - 1)(\sec^2 x)$ $= 3(\sec^4 x - \sec^2 x)$	2	2 marks • Correct solution 1 mark • Uses chain rule
7(d) (ii) $3 \int (\sec^4 x - \sec^2 x) dx = \tan^3 x$ $3 \int \sec^4 x dx = \tan^3 x + 3 \int \sec^2 x dx$ $= \tan^3 x + 3 \tan x + c$ $\int \sec^3 x dx = \frac{1}{3} \tan^3 x + \tan x + c$	2	2 marks • Correct solution 1 mark • Uses part (i) to write an expression for $\int \sec^4 x dx$

Solution		Marks	Comments
QUESTION 8			
8(a)	$\sin 3x = \sin 2x \cos x$ $= \frac{1}{2}(\sin 3x + \sin x)$ $2\sin 3x = \sin 3x + \sin x$ $\sin 3x = \sin x$ Q1, Q2 $3x = x, \pi - x$ $2x = 0 \quad \text{or} \quad 4x = \pi$ $x = 0 \quad \quad x = \frac{\pi}{4}$	4	4 marks • Correct solution 3 marks • Correctly finds one solution for x 2 marks • establishes $\sin 3x = \sin x$ • states one correct solution for x 1 mark • Uses “products to sum” formula to simplify $\sin 3x + \sin x$
8(b)	$V = \pi \int_0^{\frac{\pi}{3}} (\cos x + \sec x)^2 dx$ $= \pi \int_0^{\frac{\pi}{3}} (\cos^2 x + 2 + \sec^2 x) dx$ $= \pi \int_0^{\frac{\pi}{3}} \left\{ \frac{1}{2}(1 + \cos 2x) + 2 + \sec^2 x \right\} dx$ $= \pi \int_0^{\frac{\pi}{3}} \left(\frac{5}{2} + \frac{1}{2} \cos 2x + \sec^2 x \right) dx$ $= \pi \left[\frac{5x}{2} + \frac{1}{4} \sin 2x + \tan x \right]_0^{\frac{\pi}{3}}$ $= \pi \left(\frac{5\pi}{6} + \frac{\sqrt{3}}{8} + \sqrt{3} - 0 \right)$ $= \pi \left(\frac{5\pi}{6} + \frac{9\sqrt{3}}{8} \right) \text{ units}^3$	4	4 marks • Correct solution 3 marks • Finds the primitive function • Successfully evaluates primitive that has been obtained by correct integration of a trig function found via a relevant process 2 marks • Uses the identity $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ to rewrite integrand into an easily handled form • Finds a primitive that has been obtained by correct integration of a trig function found via a relevant process 1 mark • finds the primitive of $\sec^2 x$ <i>Note: decimal approximation of 14.34624319 is acceptable</i>
8(c) (i)	$\frac{d}{dx} \left(\tan^{-1} \frac{1-x}{1+x} \right) = \frac{\frac{(1+x)(-1) - (1-x)(1)}{(1+x)^2}}{1 + \left(\frac{1-x}{1+x} \right)^2}$ $= \frac{-2}{(1+x)^2 + (1-x)^2}$ $= \frac{-2}{2 + 2x^2}$ $= -\frac{1}{1+x^2}$	2	2 marks • Correct solution 1 mark • Correctly uses the differentiation rule for $\tan^{-1} f(x)$

Solution		Marks	Comments
QUESTION 8			
8(c) (ii)	$\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = \frac{1}{1+x^2} - \frac{1}{1+x^2}$ $= 0$ $\therefore$ as both functions are continuous for $x > -1$, $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = c$ let $x = 0$; $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \tan^{-1} 0 + \tan^{-1} 1$ $= \frac{\pi}{4}$ thus $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4}$, for $x > -1$ <i>OR</i> $\tan \left(\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = \frac{x + \frac{1-x}{1+x}}{1 - x \left(\frac{1-x}{1+x} \right)}$ $= \frac{x(1+x) + (1-x)}{(1+x) - x(1-x)}$ $= \frac{x^2 + 1}{1+x^2}$ $= 1$ $\therefore \tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \tan^{-1} 1$ $= \frac{\pi}{4}$	2	2 marks - Correct proves result 1 mark - Explains why the function is a constant - Shows that $\tan \left(\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} \right) = 1$
8(c) (iii)	let $x = \sqrt{2} - 1$; $\tan^{-1}(\sqrt{2} - 1) + \tan^{-1} \left(\frac{2 - \sqrt{2}}{\sqrt{2}} \right) = \frac{\pi}{4}$ $\tan^{-1}(\sqrt{2} - 1) + \tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{4}$ $2 \tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{4}$ $\tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{8}$	1	1 mark Correct exact value